



Comparing the Post–Earnings Announcement Drift for Surprises Calculated from Analyst and Time Series Forecasts

JOSHUA LIVNAT* AND RICHARD R. MENDENHALL†

Received 5 August 2003; accepted 19 September 2005

ABSTRACT

Post–earnings announcement drift is the tendency for a stock’s cumulative abnormal returns to drift in the direction of an earnings surprise for several weeks following an earnings announcement. We show that the drift is significantly larger when defining the earnings surprise using analysts’ forecasts and actual earnings from I/B/E/S than when using a time series model based on Compustat earnings data. Neither Compustat’s policy of restating earnings nor the inclusion of “special items” in reported earnings contribute significantly to the disparity in drift magnitudes. Rather, our results suggest that this disparity is attributable to differences between analyst forecasts and those of time-series models—or at least to factors correlated with these differences. Further, we document that analyst forecasts lead to return patterns around future earnings announcements that differ from those observed when using time-series models, suggesting that the two types of surprises may capture somewhat different forms of mispricing.

*New York University; †University of Notre Dame. The authors gratefully acknowledge the preliminary and unrestated Compustat quarterly data provided by Charter Oak Investment Systems Inc. The authors are also grateful for the contribution of Thomson Financial for providing forecast data available through the Institutional Brokers Estimate System, as part of a broad academic program to encourage earnings expectations research. The authors also thank Shai Levi, an anonymous reviewer, and the editor for their comments on earlier versions of this paper.

1. *Introduction*

Post-earnings announcement drift is the tendency for a stock's cumulative abnormal returns to drift in the direction of a recent earnings surprise for several weeks following an earnings announcement. That is, after a firm announces earnings that exceed (fall short of) a proxy for the market's expectation of earnings, subsequent abnormal returns tend to be higher (lower) than normal for several weeks or even months. This phenomenon is also called the standardized unexpected earnings (SUE) effect and has been documented consistently in numerous papers for over 35 years.¹ The purpose of this paper is to address the following questions: Are there differences in the magnitudes and patterns of abnormal returns generated in portfolios formed on competing measures of earnings surprise and, if so, what causes these differences?

We begin by showing that when the earnings surprise is defined using I/B/E/S analyst forecasts and I/B/E/S actual earnings, post-earnings announcement drift is statistically and economically larger than when using time-series models based on Compustat data. Specifically, hedge returns (between top- and bottom-surprise deciles) estimated by regressions differ by more than one percentage point per quarter between the two measures of earnings surprise. In other words, the drift is about 30% larger in magnitude when using I/B/E/S data than when using Compustat data. While at least two other papers, Affleck-Graves and Mendenhall [1992] and Doyle, Lundholm, and Soliman [2003], estimate post-earnings announcement drift using both time series and analyst forecast errors, their focuses are much different than ours. Those papers do not set out to directly compare the magnitudes or patterns of post-earnings announcement returns across different models or to investigate why differences might exist.

After we show that the SUE effect is larger when using analyst forecasts and actual earnings figures from I/B/E/S than when using time-series models based on Compustat data (the source for almost all time-series SUE studies), we investigate why this difference exists. We begin by considering deviations between the I/B/E/S and Compustat actual earnings figures.

First, Compustat follows a policy of changing firms' recorded quarterly earnings figures to reflect restated values. That is, whenever earnings are restated after an announcement (e.g., because of merger or auditor's actions) Compustat changes the originally recorded value. In contrast, I/B/E/S presumably includes the originally reported earnings in its actual earnings figures. This means that most prior drift studies use, for some observations, a Compustat earnings figure that was not the one actually seen by investors. These data errors could either inflate or understate the magnitude of the drift. In order to assess the impact of Compustat's restatement policy on

¹ A review of the post-earnings announcement drift literature appears in the next section.

the magnitude of the drift, we use a unique data set containing the original Compustat data. Specifically, adopting a procedure similar to that of Abarbanell and Lehavy [2002], we stratify our sample on the sign and magnitude of the subsequent restatements and document the magnitude of the drift using both Compustat and I/B/E/S earnings surprises for the resulting subsamples.

In addition to being restated, Compustat's earnings reflect generally accepted accounting principles (GAAP), while most analyst-tracking services report "street" measures of earnings. That is, in their reported earnings, services such as I/B/E/S exclude various expenses required by GAAP. Because Abarbanell and Lehavy [2002] show that these differences can have significant effects on tests addressing a wide range of research questions, we examine their effects on our findings.

Some expenses omitted from street earnings are included in Compustat's variable "special items" and can therefore be backed out.² Burgstahler, Jiambalvo, and Shevlin [2002] show that special items are less persistent than other earnings components. I/B/E/S's use of street earnings, which omit special items in both the forecast and the actual earnings figure, may therefore generate a better measure of surprise and thus a larger drift. For this reason, we perform an analysis nearly identical to that for restatements. Specifically, we categorize observations by the sign and magnitude of the special items embedded in Compustat earnings and document the drift for each category using both Compustat and I/B/E/S earnings surprises.

Compustat and I/B/E/S actual earnings may differ for reasons other than restatements or special items. For example, I/B/E/S observes the market reaction to the earnings announcement prior to choosing exactly which earnings components to include in street earnings. This induces a potential ex post selection bias and is another reason why I/B/E/S may generate a better measure of surprise and a larger drift. (See Abarbanell and Lehavy [2002, 2003] as examples of how different definitions of reported earnings can impact research inferences.) As with restatements and special items, we stratify observations by the sign and magnitude of the full difference between actual earnings as defined by Compustat and as defined by I/B/E/S. We then evaluate the drift for both measures of surprise within each category.

Results indicate that the relative magnitudes of the drift for the two measures of surprise may in some cases depend on the level of restatements, special items, and the general degree of similarity between Compustat and I/B/E/S actual earnings. But, more important, the same analyses indicate that these factors do not explain a significant portion of the differences

² Special items are defined by Compustat as unusual or nonrecurring items and include, among other things, losses due to fires, write-downs and write-offs, nonrecurring profits or losses on sale of assets, and investments. Bradshaw and Sloan [2002] show that most, but not all, of the adjustments in the street numbers reflect special items.

documented in the general sample. For example, we examine observations whose time-series forecast error decile is unaffected by restatements, special items, or the total difference between Compustat and I/B/E/S measures of reported earnings. For these observations, which make up between 75% and 90% of the total sample, the drift remains economically and statistically larger for the I/B/E/S-based earnings surprises than for the Compustat-based surprises. These results suggest that the larger drift associated with I/B/E/S data is not attributable to differences in reported earnings between the two sources, but rather to differences between analyst and time-series forecasts—or at least to factors associated with these differences.

Finally, we document the pattern of returns around the four earnings announcements following the surprise. That is, we replicate the tests of Bernard and Thomas [1990], who use time-series models, for our sample. We then perform similar tests after defining the earnings surprise using analyst forecasts. We show that the pattern of returns around the four subsequent announcements is different when defining surprise using analyst forecasts than when using time-series models. Specifically, when using analyst forecasts, we do not observe the reversal of returns around the fourth following earnings announcement as documented by Bernard and Thomas [1990] for time-series models. This finding contradicts Abarbanell and Bernard's [1992] interpretation of one of their tests. (We discuss these differences further in the results section.) Our results suggest that strategies using analyst forecast errors might capture a form of mispricing that differs somewhat from that represented by time-series errors. Consistent with this notion, we show that using both measures of earnings surprise leads to a significantly larger drift than using either by itself. That is, neither type of forecast error produces a drift that subsumes the other.

Our results should be of interest to researchers and practitioners alike. If researchers do not understand how the magnitude of the drift depends on the specification of earnings surprise, they stand little chance of understanding the true nature of the anomaly. For example, researchers exploring how firm-specific characteristics affect the drift's magnitude (e.g., Bhushan [1994], Bartov, Radhakrishnan, and Krinsky [2000], and Mendenhall [2004]) may be misled if they use a less-than-optimal measure of earnings surprise. Similarly, using a substandard measure of earnings surprise in tests to determine whether one or more anomalies subsume other empirical regularities (e.g., Affleck-Graves and Mendenhall [1992], Chan, Jegadeesh, and Lakonishok [1996], and Collins and Hribar [2000]) understates the drift and again could lead to erroneous conclusions. Finally, knowing the form of earnings surprise that provides the greatest drift is obviously critical to those who consider the SUE effect an example of market inefficiency and hope to exploit it.

The next section reviews the relevant literature and develops the motivation for the research question. The third section describes the sample

and the empirical methods. The next-to-last section discusses the empirical results, while the last section concludes.

2. *Literature Review and Test Motivation*

2.1 SUE LITERATURE REVIEW

Post-earnings announcement drift, also referred to as the SUE effect, is one of the best-documented and most-resilient capital markets anomalies. For example, Brennan [1991, p. 70] calls it a “most severe challenge to financial theorists,” and Fama [1998, p. 286] refers to it as “the granddaddy of all underreaction events.”

The SUE effect was first discovered by Ball and Brown [1968] using a sample extending back to the 1950s. Over the following decade, several papers using different samples and methods confirmed the drift (e.g., Jones and Litzenberger [1970], Latane, Joy, and Jones [1970], Joy, Litzenberger, and McEnally [1977], and Latané and Jones [1979]). As these studies appeared, increased interest and skepticism led to the careful, large-scale studies of Rendelman, Jones, and Latane [1982], Foster, Olsen, and Shevlin [1984], and Bernard and Thomas [1989]. Even very recent research (e.g., Francis et al. [2004], Liang [2003], Livnat [2003a, b], Mendenhall [2004], and Narayanamoorthy [2003]) continues to document the apparent slow reaction to the information contained in earnings announcements.

Bernard and Thomas [1990, p. 34] subject the drift to a diverse battery of robustness tests, such as controlling for common risk factors used in tests of the Arbitrage-Pricing Theory (APT), and conclude that much of their “evidence cannot plausibly be reconciled with arguments built on risk mismeasurement but is consistent with a delayed price response.” Given that Bernard and Thomas’s [1990] conclusion about the drift as a delayed response to the information in earnings announcements is the predominant belief among researchers, it seems that too little thought has been given to how we actually measure the earnings surprise on which the drift is based.³

All drift studies share a basic form for estimating the earnings surprise: actual earnings minus a forecast of earnings divided by a deflator. Although the choice of forecast would seem to be a critical decision, the majority of studies use a single method, almost always a time-series model, to predict earnings. Even many recent studies—including Bartov, Radhakrishnan, and Krinsky [2000], Collins and Hribar [2000], and Narayanamoorthy [2003]—use some form of time-series model to predict earnings. Only relatively recently have studies appeared that use analysts’ forecasts. These include Affleck-Graves and Mendenhall [1992], Abarbanell and Bernard [1992], Liang

³ Causes for the drift have been discussed at length in the literature. See, for example, Bernard and Thomas [1989] for a comprehensive review. Past explanations have included methodological shortcomings (e.g., Jacob, Lys, and Sabino [2000]), risk mismeasurement (e.g., Ball, Kothari, and Watts [1993]), and slow reaction to the information content of earnings (e.g., Bernard and Thomas [1990] and Ball and Bartov [1996]).

[2003], Mendenhall [2004], Francis et al. [2004] and Livnat [2003a, b]. None of these studies, however, compares the magnitude of the drift across expectational models while holding the sample constant.

2.2 THE MAGNITUDE OF THE DRIFT

A primary objective of this paper is to compare the magnitude and pattern of post-earnings announcement abnormal returns generated in portfolios formed on the basis of competing measures of earnings surprise. Of particular interest is the magnitude of the drift for Compustat time-series data versus analyst forecasts.

As stated above, three large-scale SUE studies, all using time-series forecasts, were performed in the 1980s: Rendleman, Jones, and Latané [1982] examine cumulative market-adjusted returns following earnings announcements from 1971 to 1980; Foster, Olsen, and Shevlin [1984] examine the size-adjusted drift returns between 1974 and 1981; and Bernard and Thomas [1989] subject the drift to a battery of tests over the 1974 to 1986 period. These studies consistently document a top-surprise decile versus bottom-decile drift of about 8 to 10 basis points per day—roughly 5% to 6% per quarter—for periods of approximately one quarter.

In one set of tests, Bernard and Thomas [1989, table 2] estimate a two-quarter difference in abnormal returns between top and bottom SUE deciles of 6.8%. More recently, analogous return differences of 7.1% are documented by Collins and Hribar [2000] for the 1988 to 1997 period and roughly 6% by Narayanamoorthy [2003] for the 1978 to 1998 period. Despite differences in sample period and methods, note the similarity between the magnitudes of the two-quarter drifts documented by these more recent studies and that documented by Bernard and Thomas [1989].

Although the models differ somewhat, these studies all use time-series forecasts. Generally, the magnitude of the drift for studies using analyst forecasts is very similar to that documented by papers using time-series models—roughly 8 to 10 basis points per trading day over the first quarter. For example, using Value Line data for 178 firms for 1976 to 1986, Abarbanell and Bernard [1992, figure 1, p. 1191]⁴ find post-earnings announcement hedge returns, based on top- and bottom-earnings surprise quintiles, of 4.98% and 7.02% for one- and two-quarter holding periods, respectively. Using I/B/E/S data, Liang [2003] estimates the 60-day drift to be approximately 6% between 1989 and 2000.

⁴ In a contemporaneous study, Affleck-Graves and Mendenhall [1992] examine the drift defining the surprise using both analyst and time-series forecasts. Because of their context (explaining the “Value Line Enigma”), they do not cumulate returns immediately following the earnings announcement, but rather on dates relative to Value Line timeliness rank changes. A more recent study that defines the drift using surprises formed by analyst forecasts is that of Francis et al. [2004]. Because they perform tests in calendar time, they also do not apply the drift strategy immediately following the earnings surprise. These studies probably understate the magnitude of the drift based on analyst forecasts.

Samples requiring analyst forecasts, however, by definition, omit the least well-followed firms. Prior research shows that firms with less analyst and institutional attention (Latané and Jones [1979] and Bartov et al. [2000]) and lower recent dollar trading volume (Bhushan [1994]) tend to have larger drifts. Coupled with the observation that studies using analyst forecasts and those using time-series models document drifts of similar magnitude, this suggests that, when holding the sample constant, the drift may be larger when the surprise is defined by analysts.

Finally, after completing several drafts of this paper, we became aware of Doyle, Lundholm, and Soliman [2003]. Like our study, they use analyst forecasts and time-series predictions while holding the sample constant. They examine the magnitude of the drift by estimating market-adjusted returns by surprise decile for one-, two-, and three-year periods instead of the one- or two-quarter periods of most prior studies. When earnings surprise is defined using I/B/E/S data, over the 1988 to 2000 time period, they obtain one-, two-, and three-year, hedge returns of 13.95%, 19.89%, and 23.69%, respectively. In a footnote (Doyle, Lundholm, and Soliman [2003, footnote 10, p. 13]), they state that, when defining the earnings surprise using time-series forecasts, abnormal returns for the same one-, two-, and three-year holding periods are only 3.2%, 3.5%, and 3.8%, respectively.

Our results and those presented by Doyle, Lundholm, and Soliman [2003] are generally confirmatory: the drift is larger when the earnings surprise is defined by analyst forecasts in lieu of time series predictions. But the focus of our paper is very different from that of Doyle, Lundholm, and Soliman [2003]. While Doyle, Lundholm, and Soliman [2003] scrutinize the existence and robustness of longer-term abnormal return performance, we focus on possible explanations for *why* the magnitude of the drift may differ for the two types of forecast errors. Specifically, we examine the role of Compustat's policy of restating actual earnings, the role of special items, and generally the relative importance of differences in reported earnings from different sources in defining the earnings surprise. Also, in an attempt to determine the extent to which drifts following time-series and analyst surprises represent the same or different phenomena, we compare the patterns of returns around subsequent earnings announcements. This is the motivation for the tests whose results follow.

3. *Research Design and Data*

3.1 THE COMPUSTAT DATA AND THE RESTATEMENT PROCEDURE

Most prior studies rely on earnings data as reported by Compustat for both actual earnings and time-series predictions. As discussed above, we examine several measures of earnings using Compustat's restated data and original Compustat data supplied by Charter Oak Investment Systems, Inc. (Charter Oak). We begin by describing both Compustat's restatement procedure and the Charter Oak data.

Within 2–3 days after a firm makes its preliminary earnings announcement, Compustat enters the relevant data into the quarterly database and denotes it with an update code of 2. When the firm publicly releases its Form 10-Q, or files it with the Securities and Exchange Commission, Compustat updates all available information and changes the update code to 3. The Compustat Quarterly database is further updated if a firm restates its previously reported quarterly results. For example, if a firm engages in mergers, acquisitions, or divestitures in a particular quarter and restates previously reported quarterly data to reflect these events, Compustat replaces the previously reported numbers with the restated data. Similarly, if the firm's auditor requires a firm to restate its previously reported quarterly results, for example, as part of the annual audit, Compustat updates the quarterly database to reflect the restated data.

Charter Oak collects the weekly original CD-ROM that Compustat sends to its PC clients, which contains updated data as of that week. From these weekly updates, Charter Oak has constructed a database that contains, for each firm in the Compustat Quarterly database, three numbers for each line item in each quarter. The first number is the preliminary figure that Compustat initially entered into the database—it bears an update code of 2. The second number is the “As First Reported” (AFR) figure when Compustat first changed the update code to 3 for that firm-quarter. The third number appears in the current version of Compustat, which is used by academics to construct the earnings surprise from the firm's time series.

Unlike the current Compustat database, the Charter Oak data allows us to estimate the earnings surprise as originally seen by investors. Specifically, we use the preliminary earnings figure (corresponding to the update code of 2) as the actual earnings reported in the preliminary earnings announcement on the early announcement date. We further use the AFR earnings for the same quarter in the previous year as the market expectation of earnings. In this manner, we avoid using a potentially erroneous measure of earnings surprise based on later restatements of the original data.

For example, if a firm restates its previously released quarterly earnings to reflect a subsequent merger, the current Compustat earnings figure may be larger than originally reported reflecting the earnings of the combined entities. Using the current Compustat database, therefore, introduces additional noise and potentially a hindsight bias into the surprise. If restatements tend to be systematically positively (negatively) correlated with future returns, the restatement process could cause the estimated drift to be over-(under-) stated. Since drift researchers and investors are primarily interested in extreme-surprise observations, such hindsight bias is potentially important, and the Charter Oak data can help us assess its magnitude and direction.

3.2 ESTIMATING EARNINGS SURPRISE (SUE)

Consistent with many prior studies, we define the earnings surprise (*SUE*) as actual earnings minus expected earnings, scaled by stock price. While we

estimate *SUE* using both time-series and analyst forecasts, most prior drift studies use only a time-series forecast, typically a rolling seasonal random walk (SRW) model. Consistent with these studies, our time-series measure of *SUE* is given by the following equation:

$$SUE_{jt} = \frac{(X_{jt} - X_{jt-4})}{P_{jt}} \quad (1)$$

where X_{jt} is primary Earnings Per Share (EPS) before extraordinary items for firm j in quarter t (Compustat item No. 19), and P_{jt} is the price per share for firm j at the end of quarter t from Compustat. X_{jt} and P_{jt} are unadjusted for stock splits, but X_{jt-4} is adjusted for any stock splits and stock dividends during the period $\{t-4, t\}$. As described below, if most analyst forecasts of EPS are based on diluted EPS, we use Compustat's diluted EPS figures as well in equation (1). An advantage of this definition of *SUE* is that it can be estimated for almost every firm-quarter in the Compustat database—unlike measures of earnings surprises requiring analysts' forecasts or longer time-series of earnings.⁵

To assess the effect of earnings restatements made by Compustat, we also estimate equation (1) using the preliminary earnings data from Charter Oak for X_{jt} and the AFR earnings for X_{jt-4} . This reflects the original data reported by the firm and observed by investors. Finally, we estimate equation (1) using both the current Compustat database and the Charter Oak data, after excluding "special items" from the Compustat data. Specifically, to estimate *SUE* from the Compustat data after exclusion of special items, we subtract from the primary EPS the amount of special items times 65%, divided by the number of shares used to calculate primary (or diluted, if most analysts predict diluted EPS) earnings per share (see, e.g., Bradshaw and Sloan [2002]).

We include analyses that omit special items from Compustat earnings for several reasons. First, Burgstahler, Jiambalvo, and Shevlin [2002] show that, while the market seems to underreact to special items, stock prices appear to impound relatively more of the implications of this earnings component than those of others. We would therefore like to assess how omitting special items from the earnings stream affects the profitability of the traditional *SUE* analysis. Further, as discussed above, Bradshaw and Sloan [2002] point out that analyst-tracking services such as I/B/E/S provide a street measure of actual earnings that excludes a variety of expenses. Services like I/B/E/S believe that street earnings are more directly comparable with analysts' forecasts than is Compustat's GAAP number. Because Bradshaw and Sloan [2002] point out that many, but not all, of the expenses omitted from the street number are captured by Compustat's special items variable, for completeness, we use earnings less special items as an intermediate step

⁵ Foster, Olsen, and Shevlin [1984] find that the seasonal random walk model performs as well as more complex time-series models (e.g., autoregressive integrated moving average [ARIMA] models) in predicting the drift.

between the GAAP number provided by Compustat and the street number provided by I/B/E/S.

We define *SUE* using analysts' forecasts as indicated by equation (1) after replacing the (SRW) forecast (X_{jt-4}) with a measure of analysts' expectations, that is, for comparisons between SRW and analyst forecasts, we hold reported earnings and the price deflator constant. Considering only the most recent forecast for each analyst, our measure of analysts' expectations is the median of forecasts reported to I/B/E/S in the 90 days prior to the earnings announcement. This measure of earnings surprise is similar to those used by Abarbanell and Bernard [1992], who take forecasts from Value Line, Mendenhall [2004], who deflates by the dispersion of analysts' forecasts, and Francis et al. [2004], who take forecasts from Zacks. Abarbanell and Bernard [1992] and Mendenhall [2004] take actual earnings from their forecast providers, while Francis et al. [2004] match Zacks's forecasts with Compustat earnings. As with the SRW model, we use the current and original Compustat earnings with and without special items for actual earnings. In addition, we estimate the surprise using street earnings as provided by I/B/E/S.⁶

To address the existence of outliers and nonlinearities in the earnings surprise–return relation, most drift studies classify firms into 10 portfolios based on *SUE* (see, e.g., Bernard and Thomas [1990], Bhushan [1994], and Bartov, Radhakrishnan, and Krinsky [2000]). The analysis is then performed on portfolio ranks scaled between 0 and 1. Like some other studies, we subtract 0.5 from the *SUE* decile rank in order to assign a score of 0 to a mythical median observation. The slope coefficient in the regression of abnormal returns on the *SUE* decile rank (DSUE) may be interpreted as the return to a hedge portfolio that is long on the most positive *SUE* decile and short on the most negative *SUE* decile. Because Bernard and Thomas [1990] report that the drift is insensitive to assigning firms into *SUE* deciles based on the current quarter's *SUE* values, instead of using an ex ante implementable rule such as using *SUE* cutoffs from quarter $t-1$ (see, e.g., Foster, Olsen, and Shevlin [1984]), we follow this approach.

3.3 CUMULATIVE ABNORMAL RETURNS

Daily abnormal returns are calculated as the raw daily return from the Center for Research in Security Prices (CRSP) minus the daily return on the portfolio of firms with approximately the same size (the market value of equity as of June) and book-to-market (B/M) ratio (as of the prior December). We obtain the daily returns (and cutoff points) for the size and B/M

⁶ To make proper comparisons between I/B/E/S and Compustat data, we use the unadjusted (for splits and stock dividends) I/B/E/S forecasts and actual earnings. We thus avoid the potential rounding problems pointed out by Payne and Thomas [2003]. Further, I/B/E/S determines whether most forecasts are based on primary or diluted EPS. When matching I/B/E/S forecasts and Compustat actual earnings figures, we use the earnings definition (primary or diluted EPS) as indicated by I/B/E/S.

portfolios from Professor Kenneth French's data library, based on classification of the population into six (two size and three B/M) portfolios.⁷

Our analyses require two return periods. To estimate the drift, we sum daily abnormal returns over the period from two days after the earnings announcement through one day after the following quarterly earnings announcement. To measure the immediate short-term earnings-announcement return we sum three daily abnormal returns, including the day preceding the Compustat announcement date, the announcement date, and the following day.

3.4 SAMPLE SELECTION

The analysis in this study begins with an initial sample extracted from Compustat and Charter Oak for all firms with available data over the period 1987–2003. The entire Compustat and Charter Oak sample has 1,843 firms in the first quarter of 1987, increasing gradually to 4,743 firms in the second quarter of 2003. Since most of the analysis focuses on a comparison between time-series and analyst forecasts, we require that observations for these tests have at least one analyst forecast from I/B/E/S. This subsample contains 374 firms in the first quarter of 1987, increasing to 2,431 firms in the second quarter of 2003. Other selection criteria for each observation for firm-quarter t are as follows:

- 1) The earnings announcement date is reported in Compustat for both quarter t and quarter $t+1$ (returns are cumulated through the next earnings announcement date). The earnings report dates in Compustat and in I/B/E/S (if available) differ by not more than one calendar day.
- 2) The price per share is available from Compustat as of the end of quarter t , and is greater than \$1. This reduces noise caused by small *SUE* deflators.
- 3) The market (book) value of equity at the end of quarter $t-1$ is available from Compustat and is larger than \$5 million (positive). This eliminates very small firms with low liquidity, as well as firms at their initial stages or close to liquidation.
- 4) The firm's shares are traded on the New York Stock Exchange, American Stock Exchange, or NASDAQ.
- 5) Daily returns are available in CRSP from one day before quarter t 's earnings announcement through one day after the announcement of earnings for quarter $t+1$.
- 6) Data are available to assign the firm to one of the six Fama–French portfolios based on size and B/M.
- 7) *SUE* as defined in equation (1) can be calculated for the quarter.

⁷ Data library available at <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data-library.html>.

Some tests impose additional data constraints. For example, as mentioned above, in one test, we require the earnings announcement dates for quarters $t+1$ through $t+4$ and return data from CRSP around them.

Table 1 provides summary statistics for all observations, those for which at least one analyst forecast is available, and those for which no analyst forecast is available. As can be seen in the table, for all samples, the mean and median historical *SUE* are close to zero. Note that the time-series *SUE* exhibits a wide distribution with extreme values; hence the need to transform the *SUE* into its decile ranks for the following analysis. Also note that the sample has a wide distribution of firms in terms of size (market value of equity at the end of the previous quarter) and that firms followed by analysts tend to be larger than those that are not followed. Finally, the mean earnings announcement, cumulative abnormal return (*CAR*) is small and positive (see, e.g., Penman [1984]), but is negative for the entire subsequent quarter.

4. Results

4.1 DRIFT LEVELS USING VARIOUS DEFINITIONS OF EARNINGS SURPRISES

As discussed above, we code earnings surprises into deciles, which are then adjusted by dividing the rank by nine and subtracting 0.5. When regressing the abnormal return on the adjusted earnings surprise rank, the intercept is an estimate of the return for the median observation and the slope coefficient is an estimate of the hedge portfolio return. Table 2 shows the results of these regressions for three samples: the entire sample, the subsample with at least one analyst forecast, and the subsample of Compustat firms without analyst forecasts. The table also provides statistics on estimates obtained from various definitions of earnings surprises for the same sample firms.

Table 2 shows clearly the existence of the drift for the entire sample, where the slope coefficient is equal to 0.0521, or 5.21% per quarter, significantly different from zero at significance levels below 0.0001. This drift is comparable in magnitude with those observed in prior studies, as surveyed in section 2.2 above. When special items are excluded from both actual earnings and expected earnings (EPS in the same quarter of the previous year, adjusted for stock splits), the drift is slightly larger, 5.32%, which is consistent with arguments made in the introduction regarding the difference in persistence between special items and other earnings components. This may also create an advantage for an earnings surprise that is calculated from analyst forecasts, which presumably exclude special items. When unstated earnings from Charter Oak are used, the drift (5.17%) is smaller, but by only four basis points. This suggests that Compustat's subsequent restatements have a very small effect on estimates of the drift. Presumably, if analyst forecast databases are not subsequently restated and include the original data, this factor will exhibit a very slight tendency to reduce the drift when using I/B/E/S data relative to that obtained from the current (restated) Compustat database. Finally, when special items are excluded *and* the original

TABLE 1
Summary Statistics

Variable	<i>N</i>	Mean	Std. Dev.	10th Pctl.	25th Pctl.	50th Pctl.	75th Pctl.	90th Pctl.
Panel A: All firms								
<i>SUE</i>	186064	0.002	0.764	−0.029	−0.006	0.002	0.007	0.024
<i>CAR–Subsequent Quarter</i>	186064	−0.009	0.248	−0.267	−0.131	−0.014	0.105	0.251
<i>CAR–Earnings Announcement Window</i>	186064	0.003	0.085	−0.081	−0.033	0.000	0.036	0.088
<i>Market Value of Equity (t−1)</i>	186064	1759	1051	20	49	168	688	2573
<i>Book Value of Equity (t−1)</i>	186064	575	2350	12	30	88	312	1080
<i>Price Per Share</i>	186064	25.23	525.8	3.50	7.00	14.88	26.56	41.12
Panel B: Analyst forecast subsample								
<i>SUE</i>	107893	−0.003	0.381	−0.023	−0.004	0.002	0.005	0.016
<i>SUEAF</i>	107893	−0.001	0.035	−0.006	−0.001	0.000	0.002	0.005
<i>CAR–Subsequent Quarter</i>	107893	−0.012	0.239	−0.270	−0.127	−0.009	0.105	0.240
<i>CAR–Earnings Announcement Window</i>	107893	0.002	0.083	−0.080	−0.033	0.001	0.038	0.087
<i>Market Value of Equity (t−1)</i>	107893	2823	1353	59	137	420	1423	4744
<i>Book Value of Equity (t−1)</i>	107893	876	2900	29	66	185	601	1842
<i>Price Per Share</i>	107893	27.56	445.4	6.04	11.13	20.00	32.56	47.94
<i>Number of Forecasts</i>	107893	4.764	4.685	1	2	3	6	11
Panel C: Subsample with no analyst forecasts								
<i>SUE</i>	78171	0.009	1.091	−0.038	−0.008	0.002	0.010	0.037
<i>CAR–Subsequent Quarter</i>	78171	−0.005	0.259	−0.263	−0.135	−0.020	0.104	0.268
<i>CAR–Earnings Announcement Window</i>	78171	0.003	0.089	−0.081	−0.033	−0.001	0.034	0.090
<i>Market Value of Equity (t−1)</i>	78171	291	2567	11	22	51	137	398
<i>Book Value of Equity (t−1)</i>	78171	160	1116	7	14	35	88	236
<i>Price Per Share</i>	78171	22.00	619.87	2.38	4.25	8.75	17.25	27.63

- 1) Panel A includes all firm-quarters with data to calculate *SUE* and returns during the period Q1/1987 to Q2/2003. *SUE* is calculated from the Compustat Quarterly database and is defined as earnings per share before extraordinary items (Compustat Quarterly item 19) minus (adjusted) earnings per share in the same quarter of the prior year, scaled by the price per share at the end of the quarter. Panel B includes all firm-quarters as in panel A, but where there is at least one analyst forecast during the 90-day period before the disclosure of earnings. *SUEAF* is the I/B/E/S actual minus I/B/E/S median forecast in the 90-day period before the earnings announcement date, scaled by price per share at quarter end.
- 2) Panel C includes all firm-quarters as in panel A, but where there are no analyst forecasts during the 90-day period before the disclosure of earnings.
- 3) *CAR–Subsequent Quarter* is the abnormal return on a stock, cumulated from two days after an earnings announcement through one day after the next quarterly earnings announcement. The abnormal return is the raw return minus the average return on a same size-B/M portfolio (six portfolios), as provided by Professor French.
- 4) *CAR–Earnings announcement window* is the cumulative abnormal return for the three-day window (−1,+1) centered on the earnings announcement date of the current quarter *t*.
- 5) Market (book) value of equity (in \$million) is as of the end of the previous quarter and is based on Compustat data. Price is as of quarter end.

TABLE 2
Regression of Subsequent Quarter CAR on Adjusted DSUE

	Intercept	<i>DSUE</i> <i>Earnings</i>	<i>N</i>	<i>R</i> ²
Expected sign		+		
Compustat	−0.009	0.0521	186064	0.005
Significance (<i>t</i> -statistic)	0.001	0.001		
Compustat, special items excluded	−0.009	0.0532	186064	0.005
Significance (<i>t</i> -statistic)	0.001	0.001		
Compustat original data	−0.009	0.0517	186064	0.004
Significance (<i>t</i> -statistic)	0.001	0.001		
Compustat original data, special items excluded	−0.009	0.0527	186064	0.005
Significance (<i>t</i> -statistic)	0.001	0.001		
Compustat, analyst forecasts subsample	−0.012	0.0377	107893	0.003
Significance (<i>t</i> -statistic)	0.001	0.001		
I/B/E/S actual earnings minus I/B/E/S forecast	−0.012	0.0491	107893	0.004
Significance (<i>t</i> -statistic)	0.001	0.001		
Compustat, no analyst forecasts subsample	−0.005	0.0690	78171	0.007
Significance (<i>t</i> -statistic)	0.001	0.001		

- 1) The table reports the results of regressing *CAR-Subsequent Quarter* on adjusted *DSUE*, using all firm-quarters with available data.
- 2) *SUE* is calculated as earnings minus expected earnings, scaled by price at the end of the quarter. The earnings *SUE* is transformed into a decile rank, *DSUE*, by sorting firms according to *SUE* every quarter, assigning the firms to 10 groups, and assigning the decile rank to each firm within a decile. Adjusted *DSUE* is *DSUE* divided by 9 minus 0.5.
- 3) Earnings is primary or diluted EPS before extraordinary items, depending on whether the I/B/E/S forecast is for primary or diluted EPS. EPS is unadjusted for stock splits.
- 4) Compustat uses EPS from the current Compustat Quarterly database. Compustat original data uses the preliminary (or as first reported) EPS from the Charter Oak database. When special items are excluded, special items are multiplied by 0.65 and scaled by the number of shares used to calculate EPS.
- 5) Expected earnings are either EPS for the same quarter in the prior year (adjusted for stock splits through the current quarter), or the I/B/E/S forecast, which is the mean analyst forecast of EPS during the 90-day period before the disclosure of earnings. I/B/E/S data (actual and forecast) are unadjusted for stock splits.
- 6) *CAR* is the abnormal return on a stock, cumulated from two days after an earnings announcement for quarter *t* through one day after the earnings announcement for quarter *t*+1. The abnormal return is the raw return minus the average return on a same size-B/M portfolio (six portfolios), as provided by Professor French.
- 7) A Vuong [1989] test of whether using I/B/E/S or Compustat to estimate *SUE* for the analyst forecast subsample shows that I/B/E/S-based *SUE* is superior with a test-statistic of −3.65 and significance level below 0.00013.

(unrestated) Compustat data are used, the drift is 5.27%, just slightly higher than that obtained from the current (restated) Compustat database. This estimate should use earnings that most closely resemble earnings used by I/B/E/S.

Table 2 also shows the drift for the subsample of observations with at least one analyst forecast. As is evident from the table, the current Compustat database drift for this subsample is only 3.77%, lower than the 5.21% obtained for the entire sample of firms. For these same firms, the drift that

uses both actual earnings and analyst forecasts from the I/B/E/S database is higher, 4.91%. Consistent with the literature cited above, this figure is similar in magnitude to that calculated for the entire sample using the time-series model. To test whether the models lead to significantly different drifts, footnote 7 to the table reports the Vuong [1989] test-statistic and its significance level (see Dechow [1994]). The Vuong [1989] test-statistic clearly indicates the superiority of the I/B/E/S-based *SUE* relative to the Compustat-based *SUE* for the subsample of firms with at least one analyst forecast.⁸ This disparity in drifts between I/B/E/S and Compustat *SUE* is the subject of our study. For completeness, note that when we restrict the sample to those with no analyst forecasts, the drift is larger, 6.90%, which may be consistent with lower liquidity, lower proportion of shares held by institutional investors, and greater arbitrage risk for this subsample of firms.

4.2 THE EFFECTS OF COMPUSTAT RESTATEMENTS ON DRIFT

In this section, we adopt a methodology similar to that of Abarbanell and Lehavy [2002] by sorting and categorizing observations of the analyst sample according to the sign and magnitude of the variable of interest: Compustat's restatement. Specifically, for each firm-quarter, we estimate the effect of the scaled restatement as the difference in *SUE* based on the current (restated) Compustat database and *SUE* based on the (un-restated) Charter Oak database. The latter uses the EPS from the preliminary earnings announcement for quarter t minus the AFR EPS in quarter $t-4$ (adjusted for stock splits), scaled by the price per share at the end of quarter t . Thus, the difference in *SUE* measures the combined effects of Compustat subsequent restatements on quarter t 's EPS and quarter $t-4$'s EPS, scaled by price per share. We sort firms with negative (positive) scaled restatements into three groups, plus an additional group of firms with either no restatements or restatements so small that they do not change the Compustat-based *SUE* decile. If restatements cause the superiority of the I/B/E/S-based *SUE* over the Compustat-based *SUE*, then we should observe larger drift differences for firms in the most extreme positive and/or negative restatement groups and little or no drift differences between the I/B/E/S- and Compustat-based drifts for the group with no restatements. Table 3 reports the results of the drifts for the various restatement groups.

The majority of the firm-quarters have either no restatements (80,098 observations) or restatements sufficiently small that they do not affect the

⁸ Throughout the study, when the Vuong [1989] test is applied, we also perform a regression with the same dependent variable that includes both independent variables. For this regression, we check for equality of the two slope coefficients using an F -test. For example, we perform the regression with *CAR-Subsequent Quarter* on both *DSUE* (calculated from Compustat) and *DSUEAF* (calculated from I/B/E/S). We test that the coefficients on *DSUE* and *DSUEAF* are equal. In almost all cases, where the Vuong [1989] test-statistic is significant (insignificant), the F -test is significant (insignificant) as well. When the two tests differ in their significance levels, we report it in a footnote.

TABLE 3
Regression of Subsequent Quarter CAR on Adjusted DSUE
Comparison of Compustat SUE with I/B/E/S SUE—Analyst Forecast Subsample, Various Restatement Groups

Restatement Group	Independent Variable	Restatement Price	Intercept	Drift Coefficient	N	R ²	Vuong Statistic	Sign.
Most negative	DSUE	−0.0104	−0.024 0.007	0.0473 0.059	1841	0.0019		
	DSUEAF		−0.036 0.001	0.0587 0.003	1841	0.0047	−0.748	0.227
Mild negative	DSUE	−0.0029	−0.023 0.000	0.0588 0.201	1842	0.0029		
	DSUEAF		−0.029 0.000	0.0345 0.0811	1842	0.0017	0.427	0.665
Small negative	DSUE	−0.0009	−0.015 0.004	0.0399 0.172	1842	0.0010		
	DSUEAF		−0.016 0.002	0.0637 0.002	1842	0.0051	−1.142	0.127
No significant restatements	DSUE		−0.011 0.000	0.0372 0.000	96564	0.0026		
	DSUEAF		−0.011 0.000	0.0490 0.000	96564	0.0043	−3.551	0.000
Small positive	DSUE	0.0007	−0.014 0.011	0.0651 0.028	1934	0.0025		
	DSUEAF		−0.007 0.176	0.0686 0.001	1934	0.0060	−0.863	0.194
Mild positive	DSUE	0.0024	−0.015 0.009	0.0334 0.141	1935	0.0011		
	DSUEAF		−0.011 0.041	0.0495 0.009	1935	0.0038	−0.968	0.167
Most positive	DSUE	0.0095	−0.031 0.000	0.0320 0.097	1935	0.0014		
	DSUEAF		−0.026 0.000	0.0436 0.013	1935	0.0032	−.565	0.286

- 1) The table reports the results of regressing *CAR-Subsequent Quarter* on adjusted *DSUE* or *DSUEAF*, using all firm-quarters with available data. *SUE* is calculated as earnings minus expected earnings, scaled by price at the end of the quarter. The earnings *SUE* is transformed into a decile rank, *DSUE*, by sorting firms according to *SUE* every quarter, assigning the firms to 10 groups, and assigning the decile rank to each firm within a decile. Adjusted *DSUE* is *DSUE* divided by 9 minus 0.5. Earnings is primary or diluted EPS before extraordinary items, depending on whether the I/B/E/S forecast is for primary or diluted EPS. EPS is unadjusted for stock splits. *DSUE* uses current (restated) Compustat earnings and earnings at *t*−4 as expected earnings. *DSUEAF* uses the I/B/E/S actual minus I/B/E/S median forecast in the 90-day period before the earnings announcement date.
- 2) Restatement/Price is the difference between *SUE* calculated from the current (restated) Compustat earnings minus *SUE* calculated using the Charter Oak (unrestated) Compustat database. The latter uses the preliminary EPS and the as-first-reported EPS (adjusted for splits) at quarter *t*−4.
- 3) Firms are sorted into three groups of negative (positive) restatements, as well as a group of insignificant restatements where the *DSUE* remains the same with or without restatements.
- 4) *CAR* is the abnormal return on a stock, cumulated from two days after an earnings announcement for quarter *t* through one day after the earnings announcement for quarter *t*+1. The abnormal return is the raw return minus the average return on a same-sized B/M portfolio (six portfolios), as provided by Professor French.
- 5) The Vuong [1989] test-statistic and significance level test the hypothesis that the regression model with *DSUEAF* is superior to that of *DSUE* (negative values indicate superiority of *DSUEAF*).

time-series *SUE* decile score (16,466). For this group, the I/B/E/S-based drift is 4.90%, compared with the Compustat-based drift of 3.72%. This difference is statistically significant using the Vuong [1989] test at a 0.000 level. We also do not observe any significant drift differences for the groups with the largest (positive or negative) restatements. Thus, it is unlikely that

Compustat's subsequent restatements are responsible for the greater drift predicted by I/B/E/S-based surprises.⁹

4.3 THE EFFECTS OF SPECIAL ITEMS ON DRIFT

To examine the effects of special items, we follow a procedure similar to that for restatements in the previous subsection. Specifically, we estimate the scaled effect of special items on *SUE* as the difference in *SUE* estimated from the current (and restated) Compustat database and *SUE* from the current Compustat database, after excluding the effects of special items from EPS in quarter t and quarter $t-4$. We then sort firms into three groups of both positive and negative scaled special items and a subset with either no special items (74,378 observations) or special items that are so small that they do not change the time series *SUE* decile (22,313 observations). We compare the I/B/E/S-based drift and the Compustat-based drift for each of these groups in Table 4.

As table 4 shows, for the group of observations whose time series *SUE* decile is unchanged by special items, the I/B/E/S-based drift is significantly larger than the Compustat-based drift—5.02% versus 4.21%. This difference is significant, as indicated by the Vuong [1989] test, at the 0.006 level. Thus, it does not seem likely that the exclusion of special items is responsible for the larger drift for I/B/E/S-based *SUE* for the complete sample. Nonetheless, special items do impact the magnitude of the drift for Compustat-based earnings surprises. Note that, for the most part, the drift based on time series surprises is confined to the subsets with either insignificant or slightly positive special items.¹⁰

Burgstahler, Jiambalvo, and Shevlin [2002] find that the market underreacts to special items, but by much less than it underreacts to earnings innovations in general. While the different methodologies make it difficult to compare our results with those of Burgstahler, Jiambalvo, and Shevlin [2002], our finding that the drift is greatest when the surprise is composed primarily of nonspecial items, seems broadly consistent with their findings.

4.4 THE EFFECTS OF I/B/E/S ACTUAL EPS ON DRIFT

One possible advantage of the I/B/E/S-based drift over the Compustat-based drift is that I/B/E/S officials observe the immediate market reaction

⁹ We use the preliminary earnings for time t minus the AFR earnings for $t-4$. Ideally, we would prefer to use the restated number for X_{t-4} if the restatement occurred prior to the preliminary announcement at time t . But our data do not allow us to discern exactly when restatements occurred. This alters neither the results for observations with no restatements nor, therefore, our conclusion that restatements are not responsible for differences in Compustat- and I/B/E/S-based drifts.

¹⁰ Those cases with extreme positive and negative special items may not exhibit a full range of coded decile scores. That is, when special items have very large negative (positive) effects on the time-series *SUE*, as in the bottom (top) results row, the *SUE* will tend to be negative (positive). For this reason, the best evidence regarding the impact of special items on the full sample results is probably obtained by examining the center row—where the total effect of special items has no impact on the time-series *SUE*-coded score.

TABLE 4
Regression of Subsequent Quarter CAR on Adjusted DSUE
Comparison of Compustat SUE with I/B/E/S SUE—Analyst Forecast Subsample, Various Special
Item Groups

Restatement Group	Independent Variable	Special Items /Price	Intercept	Drift Coefficient	N	R ²	Vuong Statistic	Sign.
Most negative	DSUE	−0.010	0.001	−0.0009	1942	0.000		
			0.919	0.707				
			0.003	0.0458	1942	0.000	−1.326	0.092
	DSUEAF		0.546	0.005				
Mild negative	DSUE	−0.003	−0.010	0.0130	1942	0.000		
			0.076	0.540				
			− 0.013	0.0309	1942	0.002	−0.738	0.230
	DSUEAF		0.010	0.067				
Small negative	DSUE	−0.001	−0.009	0.0278	1942	0.001		
			0.087	0.291				
			− 0.010	0.0265	1942	0.001	−0.246	0.403
	DSUEAF		0.047	0.165				
No significant special items	DSUE		− 0.013	0.0421	96691	0.003		
			0.001	0.001				
			− 0.013	0.0502	96691	0.005	− 2.534	0.006
	DSUEAF		0.001	0.001				
Small positive	DSUE	0.001	− 0.011	0.0551	1792	0.003		
			0.038	0.027				
			−0.006	0.0528	1792	0.005	−0.460	0.323
	DSUEAF		0.221	0.005				
Mild positive	DSUE	0.003	− 0.014	0.0047	1792	0.000		
			0.027	0.821				
			− 0.013	0.0173	1792	0.001	−0.427	0.335
	DSUEAF		0.010	0.306				
Most positive	DSUE	0.010	−0.010	−0.0004	1792	0.000		
			0.272	0.984				
			−0.009	0.0498	1792	0.004	−1.344	0.089
	DSUEAF		0.147	0.006				

- 1) The table reports the results of regressing *CAR—Subsequent Quarter* on adjusted *DSUE* or *DSUEAF*, using all firm-quarters with available data. *SUE* is calculated as earnings minus expected earnings, scaled by price at the end of the quarter. The earnings *SUE* is transformed into a decile rank, *DSUE*, by sorting firms according to *SUE* every quarter, assigning the firms to 10 groups, and assigning the decile rank to each firm within a decile. Adjusted *DSUE* is *DSUE* divided by 9 minus 0.5. Earnings is primary or diluted EPS before extraordinary items, depending on whether I/B/E/S forecast is for primary or diluted EPS. EPS is unadjusted for stock splits. *DSUE* uses current (restated) Compustat earnings and earnings at *t*−4 as expected earnings. *DSUEAF* uses I/B/E/S actual minus I/B/E/S median forecast in the 90-day period before the earnings announcement date.
- 2) Special Items/Price is the difference between *SUE* calculated from the current (restated) Compustat earnings minus *SUE* calculated from the current Compustat database, after excluding special items from both quarter *t* and *t*−4.
- 3) Firms are sorted into three groups of negative (positive) effects of special items, as well as a group of insignificant special items where *DSUE* remains the same if special items are excluded.
- 4) *CAR* is the abnormal return on a stock, cumulated from two days after an earnings announcement for quarter *t* through one day after the earnings announcement for quarter *t*+1. The abnormal return is the raw return minus the average return on a same-sized B/M portfolio (six portfolios), as provided by Professor French.
- 5) The Vuong [1989] test-statistic and significance level test the hypothesis that the regression model with *DSUEAF* is superior to that of *DSUE* (negative values indicate superiority of *DSUEAF*). For the most negative group, although the Vuong [1989] test is not significant, an *F*-test indicates that the coefficient on *DSUEAF* is significantly greater than that on *DSUE*, when both independent variables are included in the same regression.

to the preliminary earnings announcement before deciding on the reported earnings number to record. That is, there is no generally accepted definition of street earnings, and forecast data providers may, intentionally or unintentionally, choose a reported earnings figure that causes the surprise to exhibit a high correlation with the observed market reaction. Stated another way, I/B/E/S has a potential timing advantage in choosing reported earnings that may provide a superior (e.g., less noisy) measure of earnings surprise that investors could not observe. Under this scenario, if the drift represents underreaction to the initial earnings surprise, the I/B/E/S ex post biased surprise may generate a greater drift.

To examine this possibility, we scale the difference between I/B/E/S actual EPS and EPS from the preliminary earnings announcement as captured by Charter Oak by quarter-end price per share. We then sort observations into three groups of negative (positive) scaled differences, as well as an additional group with no differences between I/B/E/S and Compustat actual EPS. If I/B/E/S superiority is derived through the selection of a proper ex post EPS, the I/B/E/S-based drift will be superior to the Compustat-based drift in the most extreme groups, and the drifts will be approximately equal for the group with no differences in actual EPS. The I/B/E/S-based and Compustat-based drifts are compared for each group in Table 5.

As the table indicates, the superiority of the I/B/E/S-based drift over the Compustat-based drift is evident for the 80,858 observations where actual EPS is the same for I/B/E/S and Compustat. The difference for this cell is 1.23% (5.87% versus 4.64%) and the Vuong [1989] test indicates this difference is significant at the 0.000 level. The difference is not statistically different for any of the other groups, including those with the largest differences between I/B/E/S and Compustat actual EPS. Thus, it is unlikely that the superiority of the I/B/E/S-based drift over the Compustat-based drift is due to I/B/E/S's ability to choose an ex post "superior" actual EPS.

4.5 THE EFFECTS OF I/B/E/S FORECASTS ON DRIFT

The remaining difference between the I/B/E/S- and Compustat-based drifts is that the I/B/E/S-based *SUE* uses analyst forecasts to estimate the earnings surprise while the Compustat-based *SUE* uses a time-series (seasonal random walk) forecast. To investigate the effects of this difference, we use a "clean" subsample of the analyst forecast subsample. This subsample represents all observations in which the (restated) Compustat *SUE* decile rank is the same as the decile rank when both special items are excluded (from quarter t and quarter $t-4$ EPS), and the Charter Oak (unrestated) Compustat database is used to estimate *SUE*. Thus, for this clean subsample, the combined effects of Compustat subsequent restatements and special items are not sufficient to change the *SUE* decile rank. In other words, for traditional *CAR* regressions on coded scores like those of Foster, Olsen, and Shevlin [1984] and Bernard and Thomas [1989, 1990], the values of each observation's independent variables are identical to what they would

TABLE 5
Regression of Subsequent Quarter CAR on Adjusted DSUE
Comparison of Compustat SUE with I/B/E/S SUE—Analyst Forecast Subsample, Various Groups
of Actual EPS Differences

Group	Independent Variable	I/B/E/S Actual–Compustat		Drift Coefficient	N	R ²	Vuong Statistic	Sign.
		Actual /Price	Intercept					
Most negative	DSUE	−0.013	−0.009 0.063	0.0225 0.059	3192	0.0011		
	DSUEAF		−0.002 0.660	0.0246 0.021	3192	0.0017	−0.312	0.337
Mild negative	DSUE	−0.003	− 0.014 0.000	0.0656 0.000	3192	0.0101		
	DSUEAF		−0.005 0.184	0.0366 0.001	3192	0.0037	1.612	0.947
Small negative	DSUE	−0.001	−0.003 0.356	0.0289 0.020	3192	0.0017		
	DSUEAF		−0.001 0.849	0.0229 0.053	3192	0.0012	0.343	0.634
No differences in actual EPS between I/B/E/S and Compustat	DSUE		− 0.014 0.000	0.0464 0.000	80858	0.0036		
	DSUEAF		− 0.013 0.000	0.0587 0.000	80858	0.0063	− 4.088	0.000
Small positive	DSUE	0.001	− 0.014 0.000	0.0208 0.059	5817	0.0006		
	DSUEAF		− 0.015 0.000	0.0399 0.000	5817	0.0022	−1.281	0.100
Mild positive	DSUE	0.008	− 0.016 0.000	0.0158 0.151	5818	0.0004		
	DSUEAF		− 0.019 0.000	0.0100 0.353	5818	0.0001	0.369	0.644
Most positive	DSUE	0.044	0.006 0.345	0.0295 0.031	5817	0.0008		
	DSUEAF		−0.003 0.454	0.0264 0.023	5817	0.0009	−0.083	0.467

- 1) The table reports the results of regressing *CAR–Subsequent Quarter* on adjusted *DSUE* or *DSUEAF*, using all firm-quarters with available data. *SUE* is calculated as earnings minus expected earnings, scaled by price at the end of the quarter. The earnings *SUE* is transformed into a decile rank, *DSUE*, by sorting firms according to *SUE* every quarter, assigning the firms to 10 groups, and assigning the decile rank to each firm within a decile. Adjusted *DSUE* is *DSUE* divided by 9 minus 0.5. Earnings is primary or diluted EPS before extraordinary items, depending on whether the I/B/E/S forecast is for primary or diluted EPS. EPS is unadjusted for stock splits. *DSUE* uses current (restated) Compustat earnings and earnings at quarter *t*−4 as expected earnings. *DSUEAF* uses the I/B/E/S actual minus I/B/E/S median forecast in the 90-day period before the earnings announcement date.
- 2) I/B/E/S actual EPS–Compustat actual EPS is scaled by price per share at quarter end. Compustat actual EPS is the originally reported EPS in the preliminary earnings announcement as captured by Charter Oak.
- 3) Firms are sorted into three groups of negative (positive) effects of actual EPS, as well as a group of no differences in actual EPS between I/B/E/S and Compustat.
- 4) *CAR* is the abnormal return on a stock, cumulated from two days after an earnings announcement for quarter *t* through one day after the earnings announcement for quarter *t*+1. The abnormal return is the raw return minus the average return on a same-sized B/M portfolio (six portfolios), as provided by Professor French.
- 5) The Vuong [1989] test-statistic and significance level test the hypothesis that the regression model with *DSUEAF* is superior to that of *DSUE* (negative values indicate superiority of *DSUEAF*). For the mild negative group, although the Vuong [1989] test is not significant, an *F*-test indicates that the coefficient on *DSUEAF* is significantly smaller than that on *DSUE*, when both independent variables are included in the same regression.

TABLE 6
Regression of Subsequent Quarter CAR on Adjusted DSUE
Comparison of Expected Earnings—Analyst Forecast Subsample With No Significant Special
Items or Restatements

Independent Variable	Intercept	Drift	N	R ²	Vuong Statistic	Sign.
<i>DSUE</i>	-0.012	0.0401	82133	0.0030		
	0.000	0.000				
<i>DSUEAF</i>	-0.012	0.0495	82133	0.0044	-2.538	0.006
	0.000	0.000				
<i>DSUEF</i>	-0.012	0.0488	82133	0.0041	-2.326	0.010
	0.000	0.000				
<i>DSUEF</i>	-0.012	0.0468	107893	0.0039	-3.345	0.001
	0.000	0.000				

- 1) The table reports the results of regressing *CAR-Subsequent Quarter* on adjusted *DSUE*, *DSUEAF*, and *DSUEF* using all firm-quarters with available data. *SUE* is calculated as earnings minus expected earnings, scaled by price at the end of the quarter. The earnings *SUE* is transformed into a decile rank, *DSUE*, by sorting firms according to *SUE* every quarter, assigning the firms to 10 groups, and assigning the decile rank to each firm within a decile. Adjusted *DSUE* is *DSUE* divided by 9 minus 0.5. Earnings is primary or diluted EPS before extraordinary items, depending on whether the I/B/E/S forecast is for primary or diluted EPS. EPS is unadjusted for stock splits. *DSUE* uses current (restated) Compustat earnings and earnings at quarter *t*-4 as expected earnings. *DSUEAF* uses the I/B/E/S actual minus I/B/E/S median forecast in the 90-day period before the earnings announcement date. *DSUEF* uses current (restated) Compustat earnings minus the I/B/E/S median forecast in the 90-day period before the earnings announcement date.
- 2) The table is based on all observations in which the *SUE* decile rank is the same as that obtained after excluding special items from both quarter *t* and quarter *t*-4 EPS, and where the Charter Oak database is used to estimate the original (unrestated) earnings surprise. This represents a clean subsample of the analyst forecast sample in which Compustat subsequent restatements or special items are not sufficiently large to change the *SUE* decile rank.
- 3) *CAR* is the abnormal return on a stock, cumulated from two days after an earnings announcement for quarter *t* through one day after the earnings announcement for quarter *t*+1. The abnormal return is the raw return minus the average return on a same-sized B/M portfolio (six portfolios), as provided by Professor French.
- 4) The Vuong [1989] test-statistic and significance level test the hypothesis that the regression model with *DSUEAF* (*DSUEF*) is superior to that of *DSUE* (negative values indicate superiority of *DSUEAF* or *DSUEF*).
- 5) A Vuong [1989] test that the I/B/E/S-based *SUE* (*DSUEAF*) is superior to the Compustat-based *SUE* that uses I/B/E/S forecasts as expected earnings (*DSUEF*) has a test-statistic of -0.69 and significance level of 0.246.
- 6) The last two rows are based on all observations in the analyst forecast sample, without excluding firms with significant special items or restatements. It should be compared with the results for this subsample in table 2. A Vuong [1989] test that the I/B/E/S-based *SUE* (*DSUEAF*) is superior to the Compustat-based *SUE* that uses I/B/E/S forecasts as expected earnings (*DSUEF*) has a test-statistic of -1.06 and significance level of 0.145.

be if restatements and special items did not exist. Over 82,000 observations qualify for these tests. Table 6 provides the drifts for the three measures of SUE: that based on the (restated) Compustat database, *DSUE*; that based on I/B/E/S data, *DSUEAF*; and that based on the (restated) Compustat actual paired with the I/B/E/S forecast, *DSUEF*.

As table 6 shows, the I/B/E/S-based drift, *DSUEAF*, is significantly larger than the Compustat-based drift, *DSUE*, 4.95% versus 4.01%. Similarly, the drift based on *DSUEF*, which uses the Compustat actual but the I/B/E/S forecast, is significantly larger, at 4.88%, than the Compustat-based drift, and is very close in magnitude to the I/B/E/S-based drift. Footnote 5 to the table indicates that the I/B/E/S-based drift is not significantly larger than the

drift based on I/B/E/S forecasts and Compustat actual earnings. Thus, once again it appears that any ex post bias on the part of I/B/E/S, in selecting the actual EPS, is not responsible for differences in the drift. These differences are more likely related to the selection of a potentially better forecast of earnings (I/B/E/S versus seasonal random walk), or factors associated with the differences in forecasts.

4.6 AUTOCORRELATIONS OF EARNINGS SURPRISES AND MARKET REACTIONS TO FUTURE QUARTERLY ANNOUNCEMENTS

Bernard and Thomas [1990] postulate, as a potential explanation for the post-earnings announcement drift, that market expectations mimic seasonal random walk forecasts, neglecting—or underweighting (see Ball and Bartov [1996])—the autocorrelation pattern of SRW forecast errors. This would explain Bernard and Thomas's [1990] result that market returns, around the four quarters subsequent to an SRW earnings surprise, follow the same autocorrelation pattern as SRW forecast errors: positive and declining first-, second-, and third-order autocorrelations and negative fourth-order autocorrelation.

First, we attempt to determine whether the pattern of returns following SRW errors documented by Bernard and Thomas [1990] persists in a subsample of firms followed by analysts. Next, we investigate whether analyst forecast errors, which were shown earlier to generate significantly larger drifts, are followed by a similar pattern. We therefore identify all firm-quarters with a complete series of earnings surprises and returns for quarter t and the subsequent four quarters. For this subsample of 42,122 observations, we estimate the four autocorrelations of SUE ($SUEAF$), as well as six regression equations, where the independent variable is the adjusted SUE decile rank for quarter t , $DSUE_t$ ($DSUEAF_t$). The first dependent variable is the drift from two days after the earnings announcement in quarter t through one day after the earnings announcement in quarter $t+1$.¹¹ The next five dependent variables are the earnings announcement three-day cumulative abnormal returns for quarters $t+i$, where $i = 0, \dots, 4$. These results are reported in Table 7.

As the table shows, there is a significant drift over the subsequent quarter for earnings surprises from Compustat (2.54%) and, consistent with our prior findings, a larger one for earnings surprises from I/B/E/S (4.72%). The earnings response coefficient of the earnings surprise from Compustat is also lower than that from I/B/E/S, 2.79% versus 6.27%.

The autocorrelation pattern for the Compustat $DSUE$ is very similar to that observed in Bernard and Thomas [1990, panel B of table 1] with positive and declining first three autocorrelations and a large negative fourth order autocorrelation {0.416, 0.245, 0.097, and -0.167 }. The pattern of subsequent earnings announcement $CARs$, however, is very different from

¹¹ This is a repetition of prior results, which is intended to demonstrate that the differences in Compustat- and I/B/E/S-based drifts carry over to this subsample.

TABLE 7
Earnings Announcement CAR in Subsequent Quarters and SUE Autocorrelations
Comparison of I/B/E/S and Compustat—Analyst Forecast Subsample

Dependent Variable	Independent Variable	Intercept	Market Reaction	R ²	Correlations with DSUE _t	Correlations with DSUEAF _t
CAR (quarter <i>t</i> , quarter <i>t</i> +1)	DSUE _t	0.001	0.0254	0.0014		
		0.402	0.000			
	DSUEAF _t	0.002	0.0472	0.0038		
		0.397	0.000			
CAR (quarter <i>t</i>)	DSUE _t	0.004	0.0279	0.0112		
		0.000	0.000			
	DSUEAF _t	0.004	0.0627	0.0543		
		0.000	0.000			
CAR (quarter <i>t</i> +1)	DSUE _t	0.004	0.0010	0.0000	DSUE _{t-1}	0.416
		0.000	0.323			
	DSUEAF _t	0.004	0.0044	0.0003	DSUEAF _{t-1}	0.234
		0.000	0.001			
CAR (quarter <i>t</i> +2)	DSUE _t	0.003	-0.024	0.0001	DSUE _{t-2}	0.245
		0.000	0.064			
	DSUEAF _t	0.003	0.0051	0.0004	DSUEAF _{t-2}	0.156
		0.000	0.000			
CAR (quarter <i>t</i> +3)	DSUE _t	0.003	-0.0032	0.0001	DSUE _{t-3}	0.097
		0.000	0.017			
	DSUEAF _t	0.003	0.0032	0.0001	DSUEAF _{t-3}	0.127
		0.000	0.015			
CAR (quarter <i>t</i> +4)	DSUE _t	0.003	-0.0038	0.0002	DSUE _{t-4}	-0.167
		0.000	0.004			
	DSUEAF _t	0.002	0.0010	0.0000	DSUEAF _{t-4}	0.117
		0.001	0.493			

- 1) The table is based on 42,122 firm-quarters with analyst forecasts and both earnings surprises and returns for quarters *t* through *t*+4. It reports regression estimates in which the dependent variable is abnormal return and the independent variable is the adjusted *SUE* decile rank. *SUE* is calculated as earnings minus expected earnings, scaled by price at the end of the quarter. The earnings *SUE* is transformed into a decile rank, *DSUE*, by sorting firms according to *SUE* every quarter, assigning the firms to 10 groups, and assigning the decile rank to each firm within a decile. Adjusted *DSUE* is *DSUE* divided by 9 minus 0.5. Earnings is primary or diluted EPS before extraordinary items, depending on whether the I/B/E/S forecast is for primary or diluted EPS. EPS is unadjusted for stock splits. *DSUE* uses current (restated) Compustat earnings and earnings at quarter *t*-4 as expected earnings. *DSUEAF* uses the I/B/E/S actual minus I/B/E/S median forecast in the 90-day period before the earnings announcement date.
- 2) *CAR* (quarter *t*, quarter *t*+1) is the abnormal return on a stock, cumulated from two days after an earnings announcement for quarter *t* through one day after the earnings announcement for quarter *t*+1. *CAR* (quarter *t*+*i*) is the three-day cumulative return in the window (-1,+1), where day zero is the earnings announcement date for quarter *t*+*i*. The abnormal return is the raw return minus the average return on a same-sized B/M portfolio (six portfolios), as provided by Professor French.
- 3) The last two columns report autocorrelations of *DSUE*_{*t*} (*DSUEAF*_{*t*}) with *DSUE* (*DSUEAF*) at prior quarters.

theirs. For our sample, hedge returns over the subsequent four quarters are {0.0010, -0.0024, -0.0032, and -0.0038}, with only the last two associations being significantly different from zero. Bernard and Thomas [1990, panel B of table 2], on the other hand, find positive and significant hedge returns around the first two subsequent earnings announcements and an almost zero hedge return around the third following earnings announcement. Interestingly, our point estimate for the hedge return around the

fourth earnings announcement is identical to that of Bernard and Thomas [1990] (-0.38%).

To determine why our subsequent earnings announcement *CARs* differ so much from those of Bernard and Thomas [1990], we repeat the tests for our complete sample, that is, we include firms with no analyst following. In that test, we obtain hedge *CARs* of $\{0.0080, 0.0004, -0.0029, \text{ and } -0.0072\}$, with all but the second return being significant. So, when examining the full sample, our results continue to differ from those of Bernard and Thomas [1990]. Our sample does, however, exhibit the two most salient characteristics of theirs: significantly positive hedge returns around the first earnings announcement subsequent to the surprise and significantly negative hedge returns around the fourth subsequent announcement. We can only speculate that remaining differences may be due to our use of a later time period.

Table 7 also shows that autocorrelations for the I/B/E/S-based *DSUEAF* are declining and positive through four lags $\{0.234, 0.156, 0.127, \text{ and } 0.117\}$. These correlations are similar to those of Abarbanell and Bernard [1992, panel B of table 1]. Note that, like their results, the first autocorrelation is substantially smaller than the first-order autocorrelation in Compustat-based *SUE*. Note further that, also consistent with Abarbanell and Bernard [1992], the fourth-order autocorrelation is not negative, as it is in the Compustat-based *SUE*. This indicates that analyst forecasts do not suffer from a bias that assumes earnings follow a seasonal random walk. The pattern of subsequent earnings announcement *CARs* for the I/B/E/S-based *SUE*, *DSUEAF*, is $\{0.0044, 0.0051, 0.0032, \text{ and } 0.0010\}$, with the first three associations being significantly different from zero. Positive returns around the first three subsequent earnings announcements may be viewed as broadly consistent with those of Abarbanell and Bernard [1992, figure 1]. But, Abarbanell and Bernard [1992] find some evidence of a *CAR* reversal around the fourth subsequent earnings announcement.

Why might our results differ from those of Abarbanell and Bernard [1992]? First, note that the fourth subsequent return reversal documented by Abarbanell and Bernard [1992] is not statistically significant at the 0.05 level ($t = -1.89$). Second, the reversal is driven almost entirely by negative surprise firms; positive surprise firms exhibit a three-day *CAR* of -0.03% . Finally, this discrepancy could, of course, be due to differences in sample selection procedures. Abarbanell and Bernard [1992] examine about 6,400 firm-quarter observations between 1976 and 1988 and use Value Line forecasts. For this test, we examine over 42,000 observations between 1987 and 2003 and use I/B/E/S forecasts.

While we do not know for sure why our results differ from those of Abarbanell and Bernard [1992], we do know that for a large sample of firms that exhibit a significant post-earnings announcement drift, there is no reversal of returns around the fourth subsequent earnings announcement. This seems more consistent with a general pattern of underreaction than with overreliance on a mistaken earnings model such as an SRW. These results

taken with those of prior research suggest that, while some investors may exhibit overreliance on the SRW model, other types of underreaction may also exist.

4.7 DRIFT AS A FUNCTION OF BOTH MEASURES OF SURPRISE

In table 8, we present average subsequent-quarter *CARs* after sorting observations by both the time-series and analyst forecast errors. Specifically, the first five rows correspond to quintiles based on time series surprises, while the first five columns correspond to analyst surprise quintiles. The table consists, therefore, of 25 cells each reporting the average *CAR* for observations that are similar in both time-series and analyst forecast error. The

TABLE 8
CARs in Subsequent Quarters for Cells Based on Both Compustat and I/B/E/S Forecast-Error Quintiles—Analyst Forecast Subsample

		SUEAF Quintile					
		Lowest (20%)	2	3	4	Highest (20%)	All
SUE Quintile	Lowest (20%)	−0.035	−0.030	−0.025	−0.021	−0.014	−0.028
		8579	2767	1204	1844	3304	17698
		0.001	0.001	0.001	0.001	0.018	0.001
	2	−0.030	−0.031	−0.021	−0.021	0.008	−0.023
		3419	4976	3018	2395	1718	15526
		0.001	0.001	0.001	0.001	0.182	0.001
	3	−0.013	−0.021	−0.012	−0.002	0.004	−0.012
		1413	4393	5753	3254	945	15758
		0.014	0.001	0.001	0.569	0.605	0.001
	4	−0.009	−0.017	−0.014	0.003	0.011	−0.004
		1235	2640	3499	5414	2943	15731
		0.139	0.001	0.001	0.275	0.003	0.032
	Highest (20%)	−0.016	−0.024	−0.027	−0.002	0.034	0.008
		2458	2143	1595	3119	8105	17420
		0.010	0.001	0.001	0.700	0.001	0.001
	All	−0.027	−0.025	−0.017	−0.005	0.017	−0.012
		17104	16919	15069	16026	17015	82133
		0.001	0.001	0.001	0.002	0.001	0.001

- 1) The table reports the mean subsequent-quarter *CAR* by earnings surprise quintiles using firm-quarters with available data. *SUE* is calculated as earnings minus expected earnings, scaled by price at the end of the quarter. Earnings is primary or diluted EPS before extraordinary items, depending on whether the I/B/E/S forecast is for primary or diluted EPS. EPS is unadjusted for stock splits. *SUE* uses current (restated) Compustat earnings and earnings at quarter $t-4$ as expected earnings. *SUEAF* uses the I/B/E/S actual minus I/B/E/S median forecast in the 90-day period before the earnings announcement date.
- 2) The table is based on all observations in which the *SUE* decile rank is the same as that obtained after excluding special items from both quarter t and quarter $t-4$ EPS, and in which the Charter Oak database is used to estimate the original (unrestated) earnings surprise. This represents a clean subsample of the analyst forecast sample in which Compustat subsequent restatements or special items are not sufficiently large to change the *SUE* decile rank. (Although this table uses quintiles, we maintain the same definition of clean sample, and therefore the same sample, as used in table 7, above.)
- 3) *CAR* is the abnormal return on a stock, cumulated from two days after an earnings announcement for quarter t through one day after the earnings announcement for quarter $t+1$. The abnormal return is the raw return minus the average return on a same-sized B/M portfolio (six portfolios), as provided by Professor French.

bottom row presents average *CARs* by analyst forecast error quintile and the far-right column displays average *CARs* by time-series error quintile.¹²

Consistent with other results in this paper, the drift based purely on analysts is significantly greater than that based on the time-series model. The difference between the top and bottom time-series quintiles is 3.6% (0.8% – [–2.8%]), while the difference between extreme analyst forecast error quintiles is 4.4% (1.7% – [–2.7%]). More important for this table, note that neither measure of earnings surprise subsumes the other; only about 50% of the extreme (negative or positive) quintile firms are common to both *SUE* and *SUEAF*. Holding the time series quintile constant, the *CAR* for the most positive analyst forecast quintile are greater than those of the most negative quintile for all five cases. Holding the analyst forecast error constant, the same is true for four of five cases for time-series errors. Further, for the highest *SUE* (*SUEAF*) quintile but the lowest *SUEAF* (*SUE*) quintile the average *CAR* is actually negative, not positive as might be expected from its high *SUE* (*SUEAF*) score. Finally, note that the cell with the largest negative (positive) *CAR* is the cell representing both the most negative (positive) time-series and analyst forecast surprise. Comparing these two cells, each of which has over 8,000 observations, yields a drift of 6.9% (3.4% – [–3.5%])—substantially larger than that obtained using either surprise measure independently. These findings, like the autocorrelation results from the prior section, suggest the two measures of earnings surprise represent somewhat different forms of mispricing.

4.8 ROBUSTNESS

4.8.1. Control Variables. The regression of the subsequent-quarter *CAR* on the earnings surprise has been shown to be sensitive to the percentage of institutional investors (Bartov, Radhakrishnan, and Krinsky [2000]), arbitrage risk (Mendenhall [2004]), and trading volume (Bhushan [1994]). We interact these variables with the adjusted *SUE* decile rank in the regression analyses. The main results are insensitive to the inclusion of these variables.

4.8.2. Extreme Returns. We delete observations in the most extreme positive and negative 0.5% of all subsequent quarter *CARs*. This deletion does not alter inferences, but causes the significance levels in the regressions, as well as the Vuong [1989] test-statistics, to increase.

4.8.3. Firms Covered by More Than One Analyst. Since there may be significant differences between firms that are covered by only one analyst and those covered by multiple analysts, we replicate the basic tests of this paper

¹² Here we use the clean sample of table 6 described above. Each quintile is defined as two previously defined deciles. (We use quintiles here to ease exposition.) The number of observations is not equal across quintiles because deciles are defined based on surprise rank within the full sample, while the “clean” sample, as described above, uses only observations whose time-series decile is not altered by restatements or special items.

using only observations with more than one available forecast. This reduces the analyst-forecast subsample from 107,893 firm-quarter observations to 81,105 observations. Results are qualitatively identical and no inferences are altered.

5. *Conclusions*

The purpose of this paper is to assess potential differences in the magnitude and pattern of post-earnings announcement abnormal returns generated in portfolios formed on competing measures of earnings surprise. Specifically, we directly compare abnormal returns following time-series earnings surprises using Compustat data with those following analyst earnings forecast errors using I/B/E/S data.

Although the vast majority of drift studies define the earnings surprise as a time-series forecast error, we show that the drift is consistently and significantly larger when using analyst forecast errors from I/B/E/S. We further show that little, if any, of this advantage is attributable to Compustat's policy of restating quarterly earnings or to I/B/E/S's policy of excluding special items (or other types of expenses) from their definition of reported (street) earnings. The lion's share of the advantage appears to be attributable to differences between analyst and time-series forecasts or factors associated with these differences.

We also show that the pattern of returns around subsequent earnings announcements is markedly different for analyst forecast errors than that previously documented for time-series errors. Specifically, prior research documents that hedge returns (long positive surprises and short negative surprises) following a time-series forecast error are negative at the time of the fourth earnings announcement following the surprise. This has led many researchers to conclude that the apparent slow reaction to earnings announcements is due to a particular type of investor behavior: overreliance on the seasonal random walk model of earnings. For analyst forecast errors, however, which, as mentioned above, lead to a larger post-earnings announcement drift, we do not observe negative hedge returns around the fourth announcement following the surprise. This result may be viewed as suggesting that some prior explanations for the drift, for example, investor overreliance on SRW forecasts, may be premature and/or that analyst and time-series forecast errors capture somewhat different forms of stock market mispricing.

The findings that the drift is significantly larger when using analysts' forecasts and that analyst and time-series errors may not capture identical phenomena are critical to both researchers and investors. Researchers who: (1) attempt to determine the nature of the drift by exploring the firm-specific characteristics that affect its magnitude, (2) attempt to disentangle the effects of multiple capital market anomalies, and/or (3) who hope to assess the actual profitability of the drift should use the most effective form of surprise available. Investors who view the drift as a violation of market

efficiency and hope to exploit it should also use the earnings surprise signal, or combination of signals, that maximizes the drift.

REFERENCES

- ABARBANELL, J., AND V. BERNARD. "Tests of Analysts' Overreaction/Underreaction to Earnings Information as an Explanation for Anomalous Stock Price Behavior." *Journal of Finance* 47 (1992): 1181–207.
- ABARBANELL, J., AND R. LEHAVY. "Differences in Commercial Database Reported Earnings: Implications for Empirical Research." Working paper, University of North Carolina, 2002.
- ABARBANELL, J., AND R. LEHAVY. "Biased Forecasts or Biased Earnings? The Role of Reported Earnings in Explaining Apparent Bias and Over/Underreaction in Analysts' Earnings Forecasts." *Journal of Accounting and Economics* 36 (2003): 105–46.
- AFFLECK-GRAVES, J., AND R. R. MENDENHALL. "The Relation Between the Value Line Enigma and Post-Earnings-Announcement Drift." *Journal of Financial Economics* 31 (1992): 75–96.
- BALL, R., AND E. BARTOV. "How Naïve Is the Stock Market's Use of Earnings Information?" *Journal of Accounting and Economics* 21 (1996): 319–37.
- BALL, R., AND P. BROWN. "An Empirical Evaluation Of Accounting Income Numbers." *Journal of Accounting Research* 6 (1968): 159–77.
- BALL, R.; S. P. KOTHARI; AND R. L. WATTS. "Economic Determinants of the Relation Between Earnings Changes and Stock Returns." *The Accounting Review* 68 (1993): 622–38.
- BARTOV, E.; S. RADHAKRISHNAN; AND I. KRINSKY. "Investor Sophistication and Patterns in Stock Returns after Earnings Announcements." *The Accounting Review* 75 (2000): 43–63.
- BERNARD, V. L., AND J. K. THOMAS. "Post-Earnings-Announcement Drift: Delayed Price Response or Risk Premium?" *Journal of Accounting Research* 27 (Supplement 1989): 1–36.
- BERNARD, V. L., AND J. K. THOMAS. "Evidence that Stock Prices Do Not Fully Reflect the Implications of Current Earnings for Future Earnings." *Journal of Accounting and Economics* 13 (1990): 305–40.
- BHUSHAN, R. "An Informational Efficiency Perspective on the Post-Earnings Announcement Drift." *Journal of Accounting and Economics* 18 (1994): 45–65.
- BRADSHAW, M. T., AND R. G. SLOAN. "GAAP Versus The Street: An Empirical Assessment Of Two Alternative Definitions Of Earnings." *Journal of Accounting Research* 40 (2002): 41–66.
- BRENNAN, M. J. "A Perspective on Accounting and Stock Prices." *The Accounting Review* 66 (1991): 67–79.
- BURGSTAHLER, D.; J. JIAMBALVO; AND T. SHEVLIN. "Do Prices Fully Reflect the Implications of Special Items for Future Earnings?" *Journal of Accounting Research* 40 (2002): 585–612.
- CHAN, L. K. C.; N. JEGADEESH; AND J. LAKONISHOK. "Momentum Strategies." *Journal of Finance* 51 (1996): 1681–713.
- COLLINS, D. W., AND P. HRIBAR. "Earnings-Based and Accrual-Based Market Anomalies: One Effect of Two?" *Journal of Accounting and Economics* 29 (2000): 101–23.
- DECHOW, P. M. "Accounting Earnings and Cash Flows as Measures of Firm Performance: The role of Accounting Accruals." *Journal of Accounting and Economics* 18 (1994): 3–42.
- DOYLE, J.; R. LUNDHOLM; AND M. SOLIMAN. "The Extreme Future Stock Returns Following Extreme Earnings Surprises." Working paper, University of Utah, 2003.
- FAMA, E. F. "Market Efficiency, Long-Term Returns, And Behavioral Finance." *Journal of Financial Economics* 49 (1998): 283–306.
- FOSTER, G.; C. OLSEN; AND T. SHEVLIN. "Earnings Releases, Anomalies and the Behavior of Security Returns." *The Accounting Review* 59 (1984): 574–603.
- FRANCIS, J.; R. LAFOND; P. OLSSON; AND K. SCHIPPER. "Accounting Anomalies and Information Uncertainty." Working paper, Duke University, 2004.
- JACOB, J.; T. LYS; AND J. SABINO. "Autocorrelation Structure of Forecast Errors from Time-Series Models: Implications for Post-Earnings Announcement Drift Studies." *Journal of Accounting and Economics* 28 (2000): 329–58.
- JONES, C. P., AND R. H. LITZENBERGER. "Quarterly Earnings Reports and Intermediate Stock Price Trends." *Journal of Finance* 25 (1970): 143–8.

- JOY, O. M.; R. H. LITZENBERGER; AND R. W. MCENALLY. "The Adjustment of Stock Prices to Announcements of Unanticipated Changes in Quarterly Earnings." *Journal of Accounting Research* 15 (1977): 207–25.
- LATANE, H. A., AND C. P. JONES. "Standardized Unexpected Earnings—1971–1977." *Journal of Finance* 34 (1979): 717–24.
- LATANE, H. A.; O. M. JOY; AND C. P. JONES. "Quarterly Data, Sort-Rank Routines, and Security Evaluation." *The Journal of Business* 43 (1970): 427–38.
- LIANG, L. "Post-Earnings Announcement Drift and Market Participants' Information Processing Biases." *Review of Accounting Studies* 8 (2003): 321–45.
- LIVNAT, J. "Post-Earnings-Announcement Drift: The Role of Revenue Surprises and Earnings Persistence." Working paper, New York University, 2003a.
- LIVNAT, J. "Differential Persistence of Extremely Negative and Positive Earnings Surprises: Implications for the Post-Earnings-Announcement Drift." Working paper, New York University, 2003b.
- MENDENHALL, R. R. "Arbitrage Risk and Post-Earnings-Announcement Drift." *Journal of Business* 77 (2004): 875–94.
- NARAYANAMOORTHY, G. "Conservatism and Cross-Sectional Variation in the Post-Earnings-Announcement Drift." Working paper, Yale School of Management, 2003.
- PAYNE, J. L., AND W. B. THOMAS. "The Implications of Using Stock-Split Adjusted I/B/E/S Data in Empirical Research." *The Accounting Review* 78 (2003): 1049–67.
- PENMAN, S. H. "Abnormal Returns to Strategies Based on the Timing of Earnings Reports." *Journal of Accounting and Economics* 6 (1984): 165–83.
- RENDLEMAN, R. J.; C. P. JONES; AND H. A. LATANE. "Empirical Anomalies Based on Unexpected Earnings and the Importance of Risk Adjustments." *Journal of Financial Economics* 10 (1982): 269–87.
- VUONG, Q. H. "Likelihood Ratio Tests for Model Selection and Non-Nested Hypotheses." *Econometrica* 57 (1989): 307–33.